

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO4_AT2 — VISUALIZATION EVALUATION EXERCISE

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO4 – Evaluation (BL5)	Assessment:	Assessment Tool 2 – Visualization Evaluation Exercise
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO4 Statement:	<i>Evaluate the appropriateness and design choices of relationship-visualization techniques (scatterplots, correlograms, PCA, bubble/hexbin plots, and network graphs) and justify improvements.(BL5)</i>		
Student Name:	Kola Amrutha Sandhya	Reg. No.:	192524270
Section / Batch:	AIDS	Date:	24-08-26

Code of Conduct

I Kola Amrutha Sandhya 192524270 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Amrutha

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond describing the scenario: evaluate the specific strength or weakness, and justify your judgment.
- Where a question asks for a recommendation, propose a specific alternative — not just “use a different chart.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – VISUALIZATION EVALUATION EXERCISE

Rubrics for Calculation **ASSESSMENT RUBRIC**
AT2 — Visualization Evaluation Exercise

CO4 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO4-AT2 Visualization Evaluation Exercise problems, in which students are given a visualization design choice and must evaluate its appropriateness, identify limitations, and justify improvements.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant visualization principle being evaluated (e.g., overplotting, PCA variance, encoding choice, network readability).	Good understanding of the relevant principle, with only minor gaps.	Basic understanding, but with some misconceptions about which principle applies.	Poor understanding of core principles; misidentifies the nature of the issue entirely.
Explanation	25%	Clearly explains why the design choice is effective or ineffective, including its practical consequence for a reader — not just that it is 'good' or 'bad.'	Explains the evaluation with adequate reasoning, covering most of the practical consequence.	Identifies the issue but explanation is generic or only loosely tied to its practical effect.	Little or no explanation provided; the evaluation is a bare assertion.
Application	20%	Proposes a specific, well-justified recommendation or alternative that	Proposes a workable recommendation, with only minor gaps in	Proposes a recommendation, but it only partially addresses the issue or lacks	Fails to propose a workable recommendation, or it does not address the

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		directly addresses the evaluated issue.	justification.	justification.	evaluated issue.
Organization	15%	The response is clearly structured, addressing evaluation and recommendation in a logical sequence with coherent overall flow.	The response is organized, with only minor issues in flow or clarity.	Basic structure; both parts are addressed but the response lacks clarity or sequence.	Poorly organized; the response is incoherent, and one or both parts are left unaddressed.
Accuracy	10%	Both the evaluation and the recommendation are completely accurate and well-suited to the specific scenario described.	Mostly accurate, with only minor omissions or a small error.	Partially correct; the evaluation or recommendation contains notable inaccuracies.	Largely incorrect or inappropriate, with major gaps in both the evaluation and recommendation.

Scoring Guide

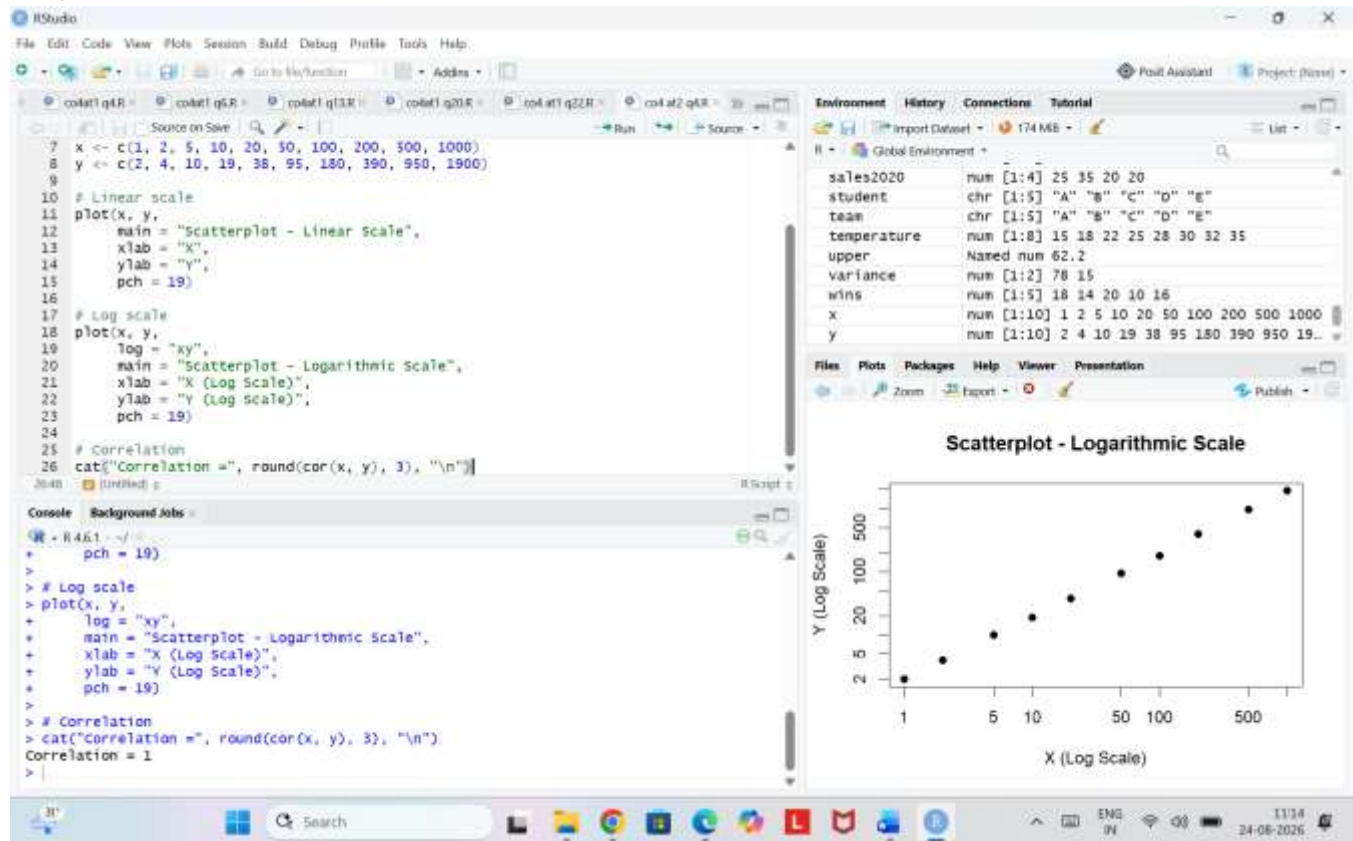
For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 4

Data: Two scatterplots show the same wide-ranging data: one uses a log-scaled axis, the other a linear axis.

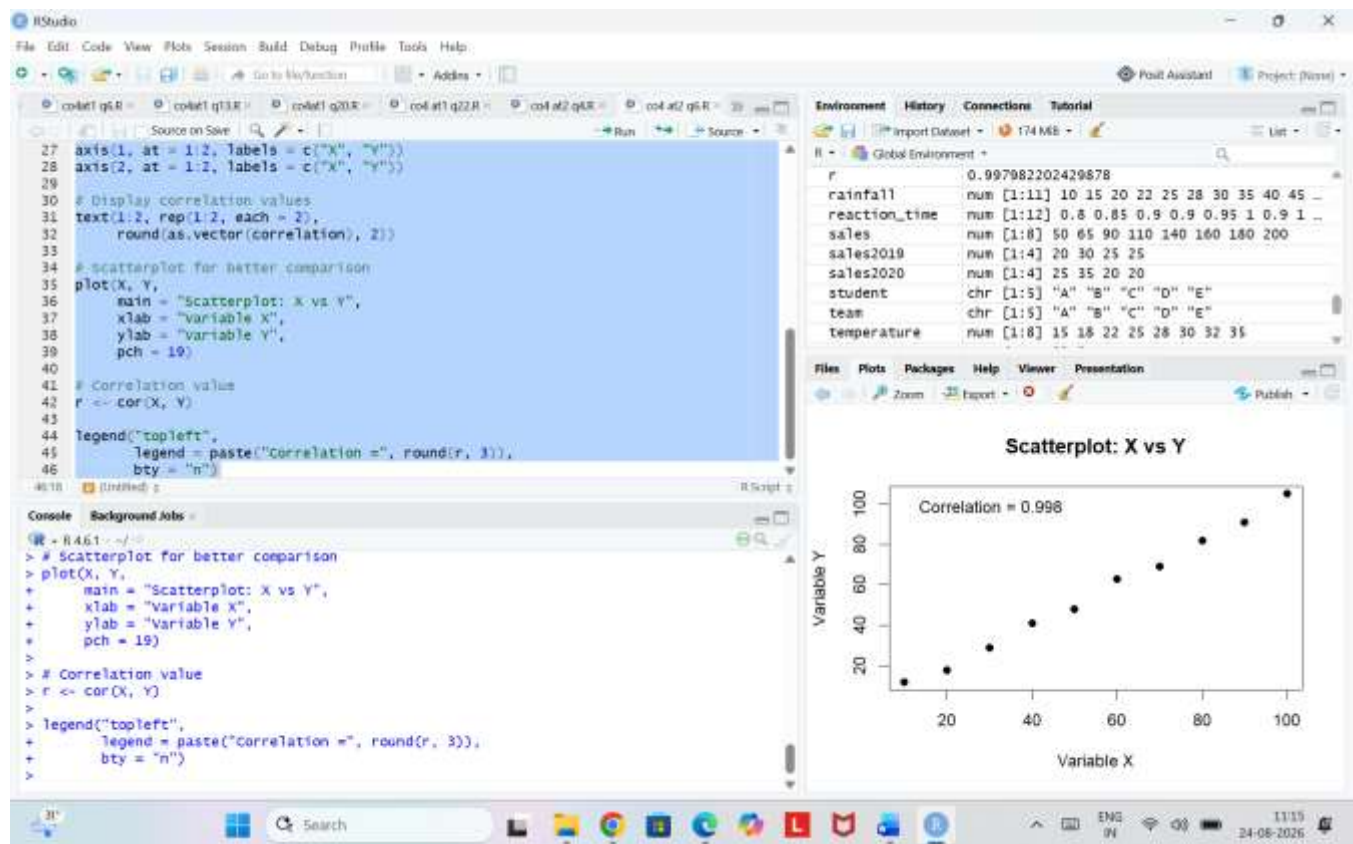
Task: Evaluate which one more accurately represents the relationship, and explain why.



Problem 6

Data: A correlogram is built using only 2 variables, producing a single off-diagonal cell.

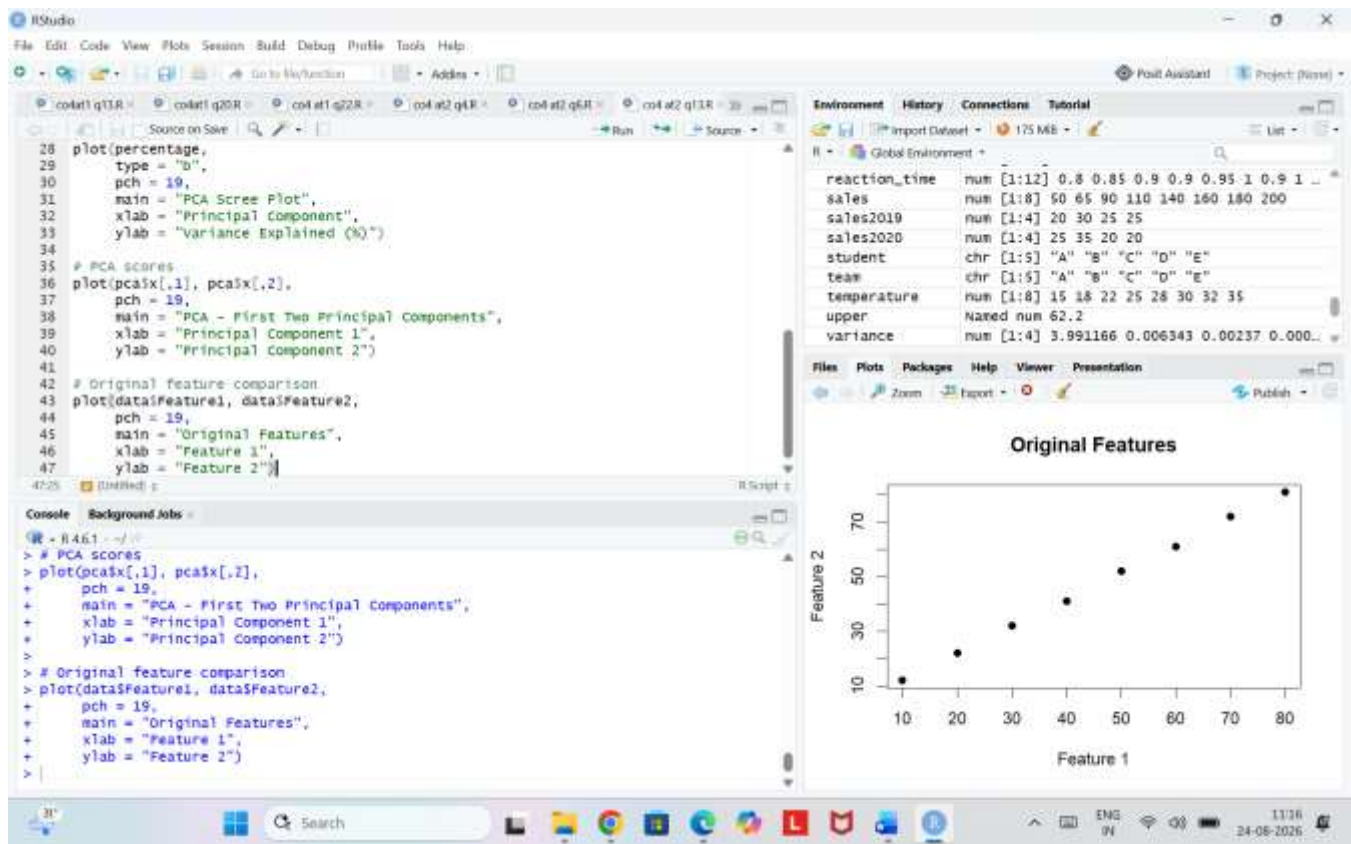
Task: Evaluate whether a correlogram is the right tool for comparing just 2 variables.



Problem 13

Data: A report claims: "PCA is always better than manually selecting the 2 most important original features."

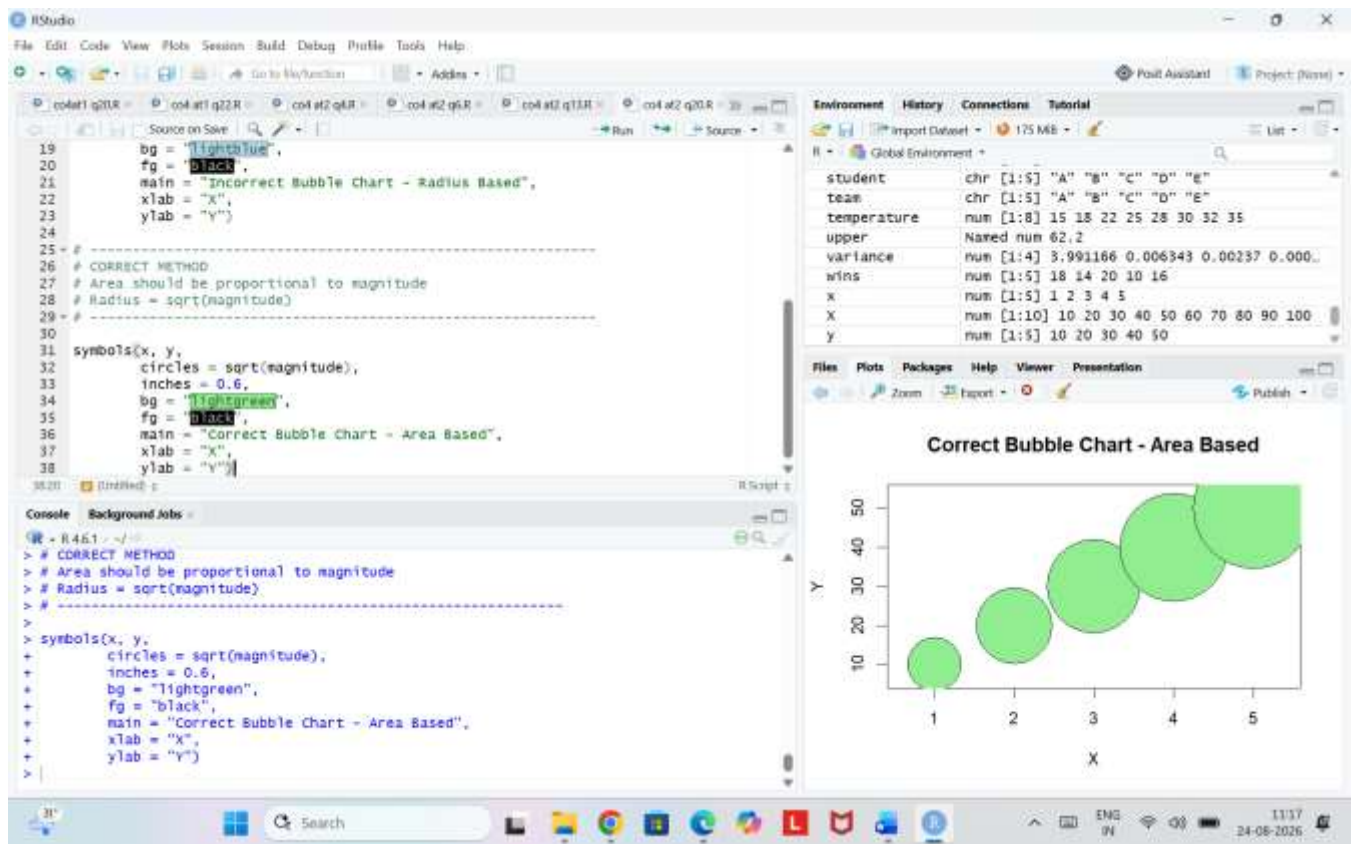
Task: Evaluate this claim.



Problem 20

Data: A report represents magnitude using bubble area, but the bubbles were actually sized using radius directly, causing the largest bubble to look disproportionately larger than justified.

Task: Evaluate this common charting mistake and explain its visual effect.



Problem 22

Data: An analyst uses node color (not size) to represent the number of connections (degree) each node has.

Task: Evaluate whether this is an effective encoding choice compared to using node size.

RStudio interface showing R code for network visualization and the resulting plot.

Source Editor:

```
32 # METHOD 1: Degree represented using COLOR
33 # -----
34
35 plot(g,
36     vertex.color = degree_values,
37     vertex.size = 25,
38     vertex.label.color = "black",
39     vertex.label.cex = 1,
40     main = "Network - Degree Represented by Color")
41
42 # -----
43 # METHOD 2: Degree represented using NODE SIZE
44 # -----
45
46 plot(g,
47     vertex.color = "lightblue",
48     vertex.size = 10 + degree_values * 8,
49     vertex.label.color = "black",
50     vertex.label.cex = 1,
51     main = "Network - Degree Represented by Node Size")
```

Environment:

Object	Class	Size
g	list of 6	
geo_data	3 obs. of 4 variables	
model	list of 12	
network	list of 10	
numeric_data	6607 obs. of 7 variables	
pca	list of 5	
reaction_data	12 obs. of 2 variables	
result	4 obs. of 4 variables	

Console:

```
> # METHOD 1: Degree represented using COLOR
> # -----
> plot(g,
+     vertex.color = "lightblue",
+     vertex.size = 10 + degree_values * 8,
+     vertex.label.color = "black",
+     vertex.label.cex = 1,
+     main = "Network - Degree Represented by Node Size")
>
```

Plot:

Network - Degree Represented by Node Size

